JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO BAGIRATHI AUTO NEEDS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

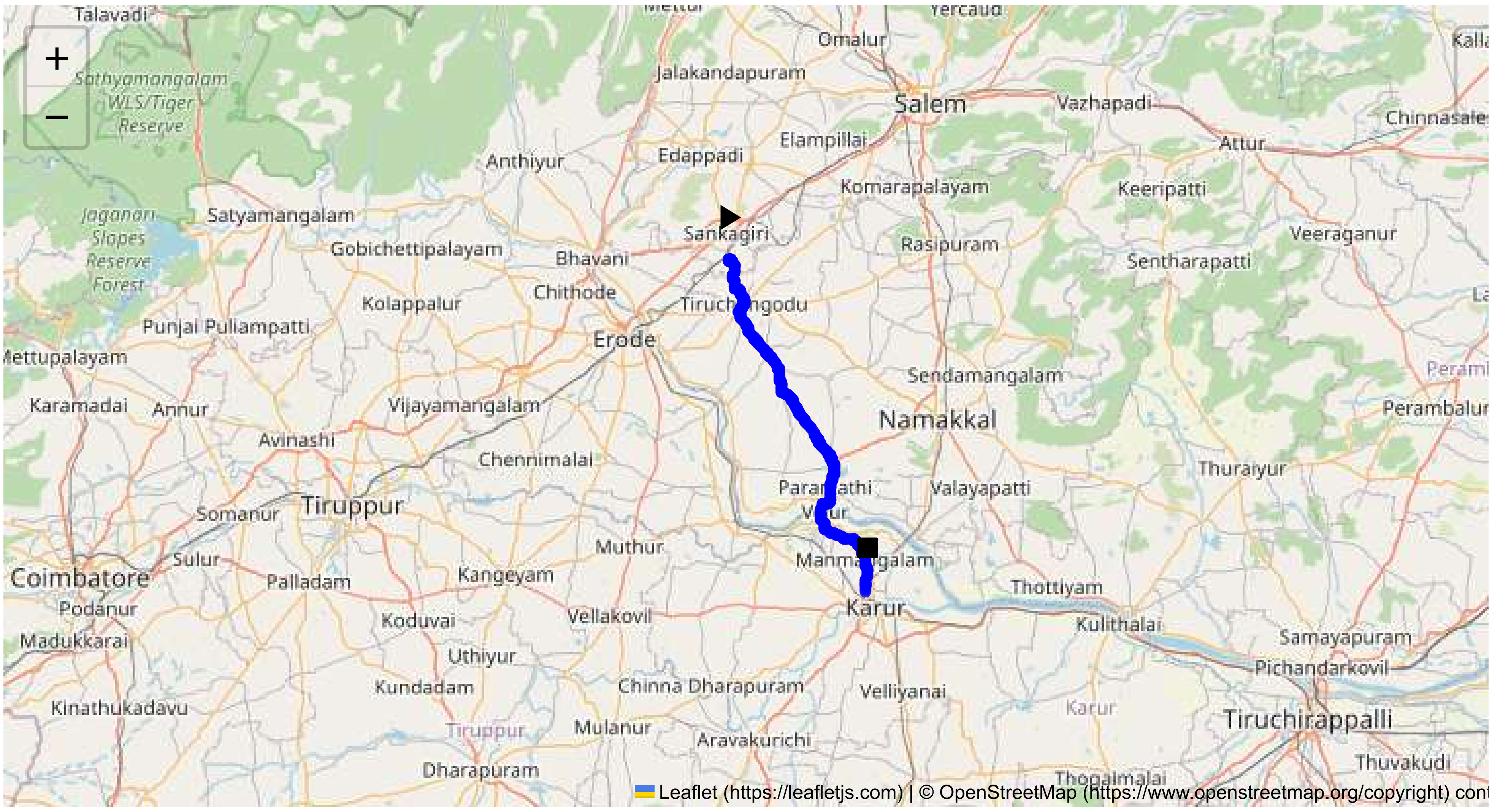
Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

Route Summary:
Total Distance: 62.13 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (10.982454, 78.064856)



Welcome to the Journey Risk Management Study

1. **Overview of the Route Map:**
   - This route spans from Sangagiri to Karur, passing through Puthur, Tiruchengode, Musalnaickenpalayam, Paramathi, and Velur. The primary highways involved include SH 86 and the Karur - Namakkal Highway. The entire journey is approximately 62.13 kilometers and features a mix of rural and semi-urban landscapes.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:**
    - Tamil Nadu experiences a tropical climate, with hot summers and modest monsoons. During the monsoon season (June to September), heavy rains can lead to waterlogging and slippery roads, particularly in rural areas. Occasional cyclones in the Bay of Bengal can also intensify during this period, causing temporary road closures.
3. **Analysis of Traffic Patterns:**
    - Traffic congestion is likely during peak hours (8-10 AM and 5-7 PM) especially near Tiruchengode and Velur junctions. These areas can be bottlenecks due to market activities and local commuting. Trucks should anticipate slower movement or consider scheduling transit outside peak hours.
4. **Assessment of Road Quality and Infrastructure:**
    - Generally, the roads on this route are well-paved, with SH 86 and the Karur-Namakkal Highway being in good condition. However, certain rural stretches between smaller villages may have narrower roads and occasional potholes, requiring careful navigation for large trucks.
5. **Suggestions for Alternative Routes for Emergencies:**
    - In case of road closure or significant delays, a viable alternative is to use the NH 44 towards Namakkal and then take the Karur bypass, although this adds significant distance. Avoid local unpaved roads which may not support heavy vehicle traffic.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:**
    - Transport of hazardous materials is subject to strict state regulations, including specific timings and designated routes in urban areas. Routes through populated areas may require additional permits, and trucks must display appropriate signage.
7. **Overview of Historical Incidents:**
    - Past incidents primarily include minor accidents due to heavy rains causing skidding or poor visibility. Instances of vehicle breakdowns in rural areas have reported delays in towing services.
8. **Environmental Considerations and Sensitive Areas:**
    - The route passes through areas with agricultural activity, demanding cautious driving to prevent spills. Additionally, local wildlife crossings in rural belts require attention, particularly at night.
9. **Analysis of Communication Coverage:**
    - Mobile network coverage is generally reliable in larger towns and along major highways. However, there may be brief dead zones in certain rural stretches between Musalnaickenpalayam and Paramathi, impacting communication.
10. **Estimated Emergency Response Times:**
    - Urban sections near Karur and Tiruchengode are equipped with quicker emergency response facilities (approximately 15-30 minutes). In contrast, rural stretches might face delays, with response times ranging from 45 minutes to an hour.
11. **An Overall Summary of Risk Assessment:**

- The Sangagiri to Karur route poses moderate risks primarily due to weather and traffic congestion. With adequate planning to avoid peak hours and adverse weather, this route is manageable for transport of hazardous materials. Consistent communication, adherence to regulations, and real-time traffic updates play key roles in safe transit.

The driver should be prepared for adverse conditions, maintain contact with control centers, and have local emergency contacts ready. Regular vehicle checks are recommended to minimize breakdown risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16755, 78.01568	30 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14480, 78.02291	15 KM/Hr
10	Blind Spot	Blind Spot	10.99285, 78.06634	10 KM/Hr
11	Blind Spot	Blind Spot	10.99286, 78.06643	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16755, 78.01568

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.99285, 78.06634

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.99286, 78.06643